

Probability Homework #9 (Coverage: 8.1, 8.2, 8.3)

- Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} k(x^2 + y^2) & \text{if } (x, y) = (1, 1), (1, 3), (2, 3) \\ 0 & \text{otherwise.} \end{cases}$$

Determine (a) the value of the constant k , (b) the marginal probability mass function of X and Y , and (c) $E(X)$ and $E(Y)$.

$$(a) \sum_{x,y \in A} p(x, y) = 2k + 10k + 13k = 1$$

$$k = \frac{1}{25} \#$$

$$(b) p_x(x) = \begin{cases} \frac{12}{25} & \text{if } x=1 \\ \frac{13}{25} & \text{if } x=2 \end{cases} \quad p_y(y) = \begin{cases} \frac{2}{25} & \text{if } y=1 \\ \frac{23}{25} & \text{if } y=3 \end{cases}$$

$$(c) E(X) = \sum_{x \in A} x \cdot p_x(x) = 1 \cdot \frac{12}{25} + 2 \cdot \frac{13}{25} = \frac{38}{25}$$

$$E(Y) = \sum_{y \in A} y \cdot p_y(y) = 1 \cdot \frac{2}{25} + 3 \cdot \frac{23}{25} = \frac{71}{25}$$

- Two fair dice are rolled. The sum of the outcome is denoted by X and the absolute value of the difference by Y . Calculate the joint probability mass function of X and Y and the marginal probability mass functions of X and Y .

$D_1 \backslash D_2$	1	2	3	4	5	6	$x \backslash y$	0	1	2	3	4	5	$p_x(x)$
1	2	3	4	5	6	7	2	$\frac{1}{36}$	0	0	0	0	0	$\frac{1}{36}$
2	3	4	5	6	7	8	3	0	$\frac{2}{36}$	0	0	0	0	$\frac{2}{36}$
3	4	5	6	7	8	9	4	$\frac{1}{36}$	0	$\frac{2}{36}$	0	0	0	$\frac{3}{36}$
4	5	6	7	8	9	10	5	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	0	$\frac{4}{36}$
5	6	7	8	9	10	11	6	$\frac{1}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{5}{36}$
6	7	8	9	10	11	12	7	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	$\frac{6}{36}$
							8	$\frac{1}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{5}{36}$
							9	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	0	$\frac{4}{36}$
							10	$\frac{1}{36}$	0	$\frac{2}{36}$	0	0	0	$\frac{3}{36}$
							11	0	$\frac{2}{36}$	0	0	0	0	$\frac{2}{36}$
							12	$\frac{1}{36}$	0	0	0	0	0	$\frac{1}{36}$
							$p_y(y)$	$\frac{6}{36}$						

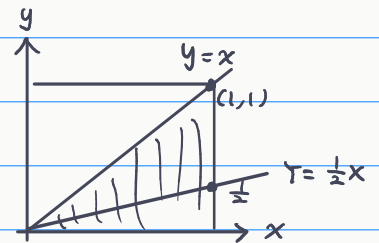
3. Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} 2 & \text{if } 0 \leq y \leq x \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Calculate the marginal probability density functions of X and Y , respectively.
 (b) Find $E(X)$ and $E(Y)$.
 (c) Calculate $P(X < 1/2)$, $P(X < 2Y)$, and $P(X = Y)$.
 (d) Are X and Y independent? Why or why not?
 (e) $f_{X|Y}(x|y) = ?$

$$(a) f_X(x) = \int_0^x 2 dy = 2x$$

$$f_Y(y) = \int_y^1 2 dx = 2 - 2y$$



$$(b) E(X) = \int_0^1 \int_0^x x \cdot 2 dy dx = \int_0^1 2x^2 dx = \frac{2}{3}$$

$$E(Y) = \int_0^1 \int_y^1 y \cdot 2 dx dy = \int_0^1 2y - 2y^2 dy = \frac{1}{3}$$

$$(c) P(X < \frac{1}{2}) = \int_0^{\frac{1}{2}} f_X(x) dx = x^2 \Big|_0^{\frac{1}{2}} = \frac{1}{4} \#$$

$$P(X < 2Y) : \text{面積法} \quad \text{Total} = 1 \times 1 \times \frac{1}{2} = \frac{1}{2} \quad \text{Area} = 1 \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$\frac{1}{4} \div \frac{1}{2} = \frac{1}{2} \#$$

$$P(X < 2Y) : \int_0^1 \int_{\frac{x}{2}}^x 2 dy dx = \int_0^1 x dx = \frac{1}{2} x^2 \Big|_0^1 = \frac{1}{2}$$

$$P(X=Y) : \text{一條線} = 0$$

$$(d) \text{ No, since } f_{X,Y} \neq f_X(x) \cdot f_Y(y)$$

$$(e) f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{2}{2-2y} \quad \text{if } 0 \leq y \leq x \leq 1$$

4. Let the joint probability mass function of random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{7}x^2y & \text{if } (x, y) = (1, 1), (1, 2), (2, 1) \\ 0 & \text{elsewhere.} \end{cases}$$

Are X and Y independent? Why or why not?

$$p_X(x) = \begin{cases} \frac{3}{7} & \text{if } x=1 \\ \frac{4}{7} & \text{if } x=2 \end{cases} \quad p_Y(y) = \begin{cases} \frac{5}{7} & \text{if } y=1 \\ \frac{2}{7} & \text{if } y=2 \end{cases}$$

$$p(1, 1) = \frac{1}{7} \neq p_X(1) \cdot p_Y(1) = \frac{3}{7} \cdot \frac{5}{7} = \frac{15}{49}, \text{ so No}$$

5. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} 2e^{-(x+2y)} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X^2Y)$.

$$h(x, y) = x^2 \cdot y$$

$$E(h(x, y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) \cdot f(x, y) \, dx \, dy$$

$$= \int_0^{\infty} \int_0^{\infty} x^2 y \cdot 2e^{-(x+2y)} \, dx \, dy$$

$$= \int_0^{\infty} 2y \int_0^{\infty} x^2 \cdot e^{-(x+2y)} \, dx \, dy$$

$$= \int_0^{\infty} 2y \cdot e^{-(x+2y)} \cdot (-x^2 - 2x - 2) \Big|_0^{\infty} \, dy$$

$$= \int_0^{\infty} 2y \cdot e^{-2y} \cdot 2 \, dy$$

$$= 4 \int_0^{\infty} ye^{-2y} \, dy$$

$$= 4 \cdot e^{-2y} \cdot \left(-\frac{1}{2}y - \frac{1}{4}\right) \Big|_0^{\infty}$$

$$= 4 \cdot \frac{1}{4} = 1$$

$$\begin{array}{l} \text{積} \quad \text{微} \\ e^{(2y-x)} \quad x^2 \\ -e^{2y-x} \quad 2x \\ e^{2y-x} \quad 2 \\ -e^{2y-x} \quad 0 \\ \hline \text{積} \quad \text{微} \\ e^{-2y} \quad y \\ -\frac{1}{2}e^{-2y} \quad 1 \\ \frac{1}{4}e^{-2y} \quad 0 \end{array}$$

6. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{25}(x^2 + y^2) & \text{if } x = 1, 2, \quad y = 0, 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $p_{X|Y}(x|y)$, $P(X = 2 | Y = 1)$, and $E(X | Y = 1)$.

$$p(x|y) = \frac{p(x, y)}{p_Y(y)} = \frac{x^2 + y^2}{5 + 2y^2} \quad \# \quad x=1, 2 \quad y=0, 1, 2$$

$$p_Y(y) = \sum_{x \in A} p(x, y) = \frac{1}{25} (1 + y^2 + 4 + y^2) = \frac{1}{25} (5 + 2y^2)$$

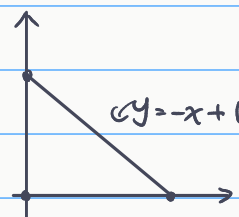
$$P(X=2|Y=1) = p(2, 1) = \frac{5}{7} \quad \#$$

$$\begin{aligned} E(X|Y=1) &= \sum_{x \in A} p(x|1) \cdot x \\ &= 1 \cdot p(1, 1) + 2 \cdot p(2, 1) \\ &= \frac{2}{7} + \frac{10}{7} = \frac{12}{7} \quad \# \end{aligned}$$

7. A point (X, Y) is selected randomly from the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.

- Find the joint probability density function of X and Y .
- Calculate $f_{X|Y}(x|y)$.
- Evaluate $E(X | Y = y)$.

$$\begin{aligned} \text{(a) Area of triangle} &= \frac{1}{2} \\ f(x, y) &= \begin{cases} \frac{1}{2} & \text{if } x \geq 0, y \geq 0, y < -x + 1 \\ 0 & \text{otherwise} \end{cases} \quad \# \end{aligned}$$

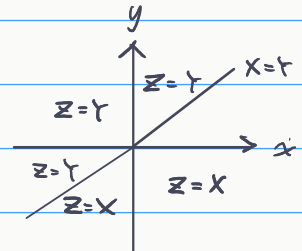


$$\begin{aligned} \text{(b) } f_Y(y) &= \int_0^{-y+1} \frac{1}{2} \cdot dx = \frac{1}{2} x \Big|_0^{-y+1} = \frac{1}{2} (1-y) \\ f(x|y) &= \frac{f(x, y)}{f_Y(y)} = \frac{1}{1-y} \quad \text{if } x \geq 0, y \geq 0, y < -x + 1 \quad \# \end{aligned}$$

$$\begin{aligned} \text{(c) } E(X|Y=y) &= \int_0^{-y+1} x \cdot \frac{1}{1-y} dx = \frac{1}{1-y} \int_0^{-y+1} \frac{1}{2} x^2 \Big|_0^{-y+1} \\ &= \frac{1}{1-y} \cdot \frac{1}{2} \cdot (1-y)^2 = \frac{1-y}{2} \quad \# \end{aligned}$$

8. Let X and Y be two independent (continuous) uniform random variables chosen from the interval $[-1, 1]$, respectively. Let $Z = \max(X, Y)$. Then (a) find the cumulative distribution function of Z , $F_Z(t) = ?$ (b) compute the expected value of Z , $E(Z)$.

$$\begin{aligned}
 F_Z(t) &= P(\max(X, Y) \leq t) = P(X \leq t, Y \leq t) \\
 &= \int_{-1}^t f(x) dx \cdot \int_{-1}^t f(y) dy \\
 &= \int_{-1}^t \frac{1}{2} dx \cdot \int_{-1}^t \frac{1}{2} dy \\
 &= \left(\frac{1}{2}t + \frac{1}{2}\right) \cdot \left(\frac{1}{2}t + \frac{1}{2}\right) \\
 &= \frac{1}{4}t^2 + \frac{1}{2}t + \frac{1}{4} \quad \#
 \end{aligned}$$



$$f_Z(t) = \frac{1}{2}t + \frac{1}{2}$$

$$\begin{aligned}
 E(Z) &= \int_{-1}^1 z \cdot f_Z(z) dz \\
 &= \int_{-1}^1 z \cdot \left(\frac{1}{2}z + \frac{1}{2}\right) dz \\
 &= \int_{-1}^1 \left(\frac{1}{2}z^2 + \frac{1}{2}z\right) dz \\
 &= \left[\frac{1}{6}z^3 + \frac{1}{4}z^2\right]_{-1}^1 \\
 &= \frac{1}{6} + \frac{1}{4} - \left(-\frac{1}{6} + \frac{1}{4}\right) \\
 &= \frac{1}{3}
 \end{aligned}$$